

Build Your Own Rule-Based Python Chatbot

30-Minute Student Task • Project Assessment

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Programming Language	Python
Project Type	Rule-Based Conversational Chatbot

Assessment

A simple rule-based conversational chatbot was developed in Python. The chatbot accepts user input continuously, recognizes greetings and selected questions, responds according to predefined rules, and exits when the user types **bye**, **exit**, or **quit**.

Key Features

- Accepts the user's name at the start of the conversation.
- Normalizes input using `strip()` and `lower()` for easier rule matching.
- Responds to greetings such as hello, hi, and hey.
- Handles the question 'how are you' and provides a help response.
- Provides a fallback response when the input does not match a rule.
- Ends the conversation when an exit keyword is entered.

The implementation and terminal demonstration are presented on the following pages.

Python Chatbot Implementation

The following is the Python implementation used for the rule-based chatbot.

```
def rule_based_bot():
    name = input("What is your name? ")

    print(f"\nBot: Hello {name}! I am your Python chatbot.")
    print("Bot: Type 'bye', 'exit', or 'quit' to end the conversation.\n")

    while True:
        user_input = input("You: ").strip().lower()

        if user_input in ["bye", "exit", "quit"]:
            print(f"Bot: Goodbye {name}! Have a great day!")
            break

        elif "hello" in user_input or "hi" in user_input or "hey" in user_input:
            print(f"Bot: Hello {name}! How can I help you today?")

        elif "how are you" in user_input:
            print("Bot: I am doing great, thank you! How about you?")

        elif "your name" in user_input:
            print("Bot: I am a simple Python rule-based chatbot.")

        elif "help" in user_input:
            print("Bot: I can respond to greetings, tell you about myself,")
            print("      answer how I am doing, and help you with basic questions.")

        else:
            print("Bot: I'm sorry, I don't understand that.")
            print("Bot: Type 'help' to see what I can do.")

if __name__ == "__main__":
    rule_based_bot()
```

How the Program Works

The function first asks for the user's name and prints an introductory message. Inside an infinite while loop, each message is converted to lowercase and stripped of extra spaces. A sequence of if/elif rules checks the message for exit commands, greetings, common questions, and help requests. If no rule matches, the chatbot uses a default response.

Core logic: Input → Normalize → Match rule → Generate response → Continue or exit.

Chatbot Execution and Test Output

Terminal-style demonstration of the chatbot responding to user inputs.

```
What is your name? Pavan

Bot: Hello Pavan! I am your Python chatbot.
Bot: Type 'bye', 'exit', or 'quit' to end the conversation.

You: hello
Bot: Hello Pavan! How can I help you today?

You: how are you
Bot: I am doing great, thank you! How about you?

You: yes can help me give the roadmap of python
Bot: I can respond to greetings, tell you about myself,
    answer how I am doing, and help you with basic questions.

You: how can u help me
Bot: I can respond to greetings, tell you about myself,
    answer how I am doing, and help you with basic questions.

You: what is ur name
Bot: I'm sorry, I don't understand that.
Bot: Type 'help' to see what I can do.

You: hello
Bot: Hello Pavan! How can I help you today?

You: how are u
Bot: I'm sorry, I don't understand that.
Bot: Type 'help' to see what I can do.
```

Conclusion

This project demonstrates the basic structure of a rule-based chatbot using Python. It shows how conditional statements, string processing, loops, user input, and formatted output can be combined to create a simple conversational program. The chatbot is deterministic: its response depends on the predefined rules in the program.

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